lignified walls. Their ends are tapering, blunt or chisel-like and devoid of protoplast. They have large lumen without any content. Their function is conduction of water and providing mechanical support to the plant.

Xylem fibres: These cells are elongated, lignified and pointed at both the ends. Xylem fibres provide mechanical support to the plant.

Xylem vessels: These are long cylindrical, tube like structures with lignified walls and wide central lumen. These cells are dead as these do not have protoplast. They are arranged in longitudinal series in which the partitioned walls (transverse walls) are perforated, and so the entire structure looks-like a water pipe. Their main function is to transport of water and also to provide mechanical strength.

Xylem parenchyma: These are living and thin walled cells. The main function of xylem parenchyma is to store starch and fatty substances.

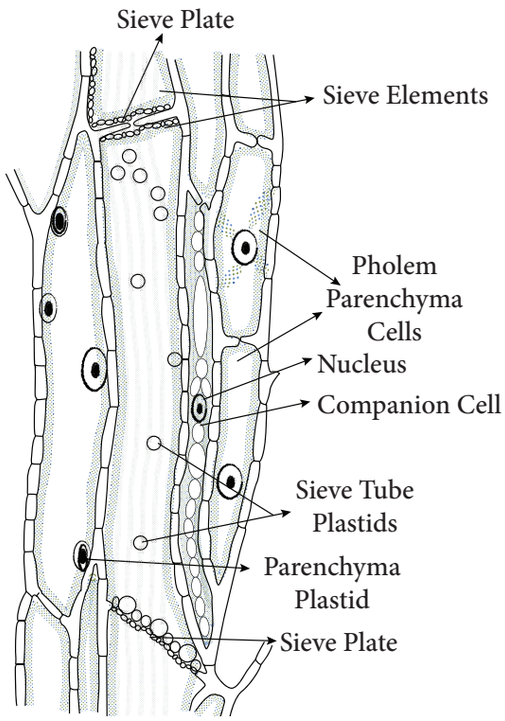


Figure 18.6 Longitudinal section of Phloem tissue

ii. Phloem

Phloem is a complex tissue and consists of the following elements: Sieve elements, Companion cells, Phloem fibres, and Phloem parenchyma.

Sieve elements: The conducting elements of phloem are collectively called as Sieve elements. Sieve tubes are elongated, tube-like slender cells placed end to end. The transverse walls at the ends are perforated and are known as sieve plates. The main function of sieve tubes is translocation of food, from leaves to the storage organs of the plants.

Companion cells: These are elongated cells attached to the lateral wall of the sieve tubes. A companion cell may be equal in length to the accompanying sieve tube element or the mother cell may be divided transversely forming a series of companion cells.

Phloem parenchyma: The phloem parenchyma are living cells which have cytoplasm and nucleus. Their function is to store food materials.

Table 18.1 Differences between Xylem and Phloem.

Xylem	Phloem
Conducts water and minerals.	Conducts organic solutes or food materials.
Conduction is mostly unidirectional i.e., from roots to apical parts of the plant.	Conduction may be bidirectional from leaves to storage organs and growing parts or from storage organs to growing parts of plants.
Conducting channels are tracheids and vessels.	Conducting channels are sieve tubes.
Component of xylem include tracheid vessels, xylem parenchyma and xylem fibres.	Components are sieve elements, companion cells, phloem parenchyma and phloem fibres.

IX_SCI_EM_Unit-18_BIO.indd 213 06-02-2020 13:10:25